# Milestone 2: Project Report
# How is Startup Success Determined?
# Group 1

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# Business Background

For this group project, we chose the Global Startup dataset from Kaggle. This dataset provides a comprehensive view of global startups across multiple industries, tracking their early-stage revenue growth, founder details, geographic distribution, and final outcomes. It includes a mix of well-known real startups (e.g., SpaceX, Stripe, Airbnb, Zomato) and synthetic but realistic startup profiles designed to mirror real-world startup behavior. The data is structured to focus on early-stage startup performance.

To be more specific, the dataset contains over 500 startup records having global coverage in over 30 countries, which spans multiple industries such as AI/ML, FinTech, HealthTech, SaaS, BioTech, EdTech, E-Commerce Logistics, AgriTech, CleanTech, Cybersecurity and PropTech. It has a 3-year revenue tracking column per startup and the status of the startup based on its current operational performance.

Therefore, our main story focus was to find out how startup success was determined through unique data visualizations. For this, we wanted to find the key startup success indicators which would help us explain the various trends and patterns in our visualizations.

# Problems Identified

Our target audience is Venture Capitalist Company Executives (e.g. Andreessen Horowitz, Sequoia Capital, Accel, and General Catalyst) who want to invest in startups with a high Return on Investment (ROI). For that, they need to know how to measure startup success for effective investment decisions. As a result, we wrote a set of critical questions that will help the Venture Capitalist Company Executive instantly recognize good startup performance that are likely to succeed and yield a high ROI. The critical questions answered by our visualizations are:

- What business features maximize short-term revenue growth?
- Which founders have the highest short-term revenue?
- Which locations have the highest concentration of startups by current status, and which have the highest average year 3 revenue?
- Which regions have the most successful startups?
- How do startup revenues scale over time?
- Which founding year contains the most successful start-ups?

# Analysis

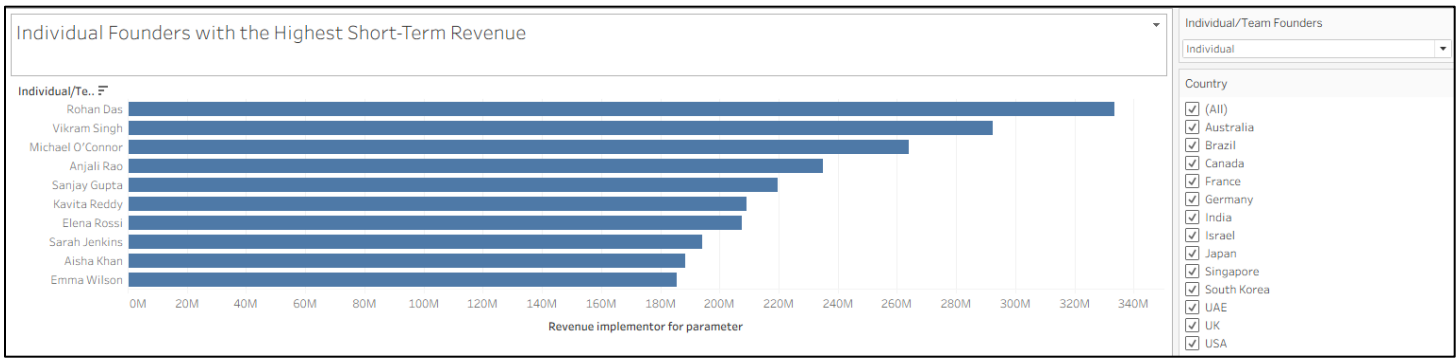
Based on the problems identified, our main goal is to find key startup success indicators that will help Venture Capitalist Company Executives make effective investment decisions so they can

invest in startups with a higher ROI. Each of these visualizations will help the Venture Capitalist Company Executives get a better understanding of how the launch dates, growth rates, countries, regions, and founder influences determine startup success.

# Data Analysis

## Visualization 1: Horizontal Bar Chart

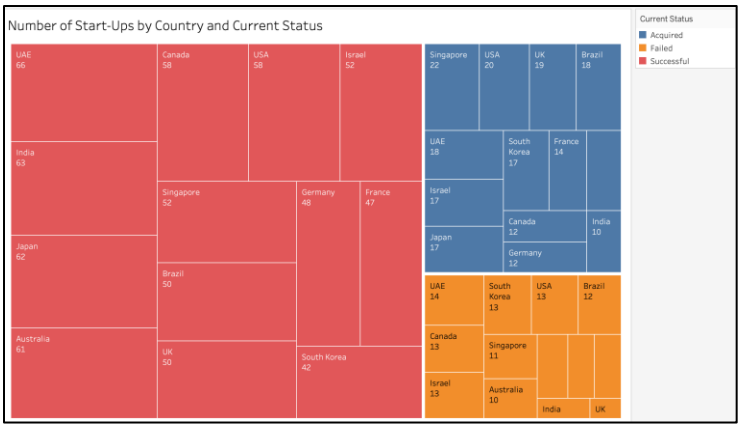
**Focus:** Founders with the Highest Short-Term Revenue.




To capture founder credibility, a horizontal bar chart showing short-term revenue for each founder was created. Founder credibility is incredibly important to capture because not all of our clients will have extensive industry knowledge in the business they want to invest in. This chart will explain which founders to watch for, whether they individually found a business or work in a team. Both of these scenarios are visualized through the use of a parameter that can switch between team founders and individual founders to show the top ten in the chosen category. To push this further, it is also possible to choose which country to filter by, which is important because of laws regarding the founding and expansion of a business. This chart can accurately represent founder credibility by proving that the people behind the wheel of the business are competent, not only in their field but also their environment.

## Visualization 2: Treemap

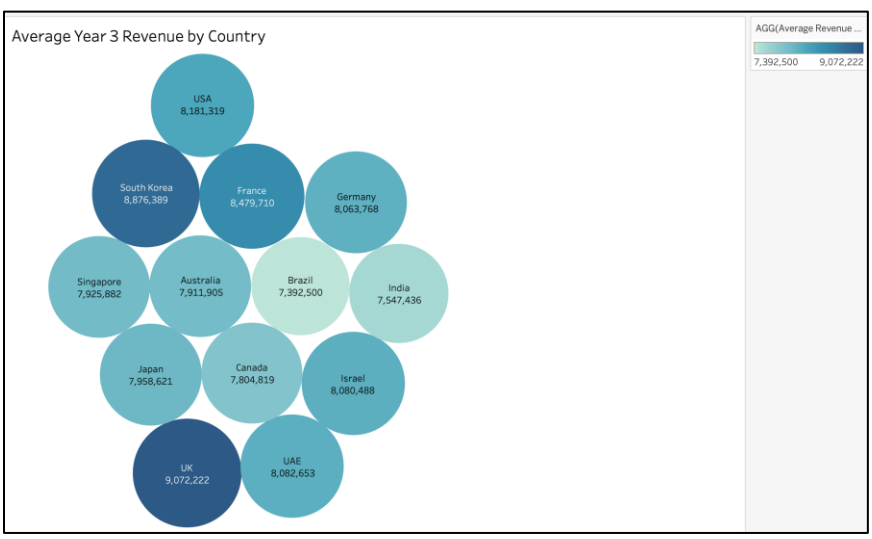
**Focus:** Number of Startups by Country and Current Status.




To identify where startups succeed most, we built a treemap where rectangle size represents startup count per country, and color indicates the operational status of the startup, where red is for Successful, blue for Acquired, orange for Failed. The dominant red tells us that successful startups significantly outnumber failed ones across this dataset. Drilling into the successful segment, UAE, India, and Japan emerge as the clear leaders. This comes down to four structural advantages these countries share: large domestic markets that absorb startup products at scale, active government diversification programs that de-risk early investment, strong corporate ecosystems that provide co-investment and exit pathways, and deep talent pipelines that reduce execution risk. For a venture capitalist, startups in these three countries carry structurally lower risk and higher scaling potential, which makes them priority candidates for investment.

## Visualization 3: Bubble Chart

**Focus:** Average Year 3 Revenue by Country.

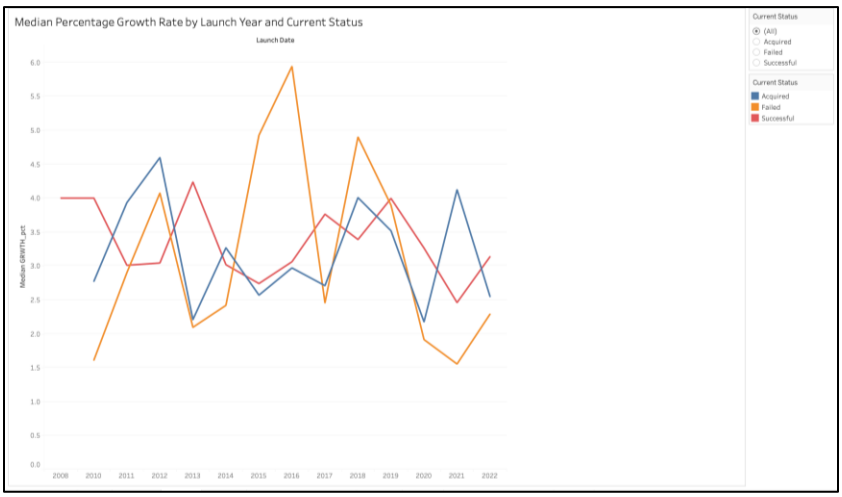



To identify which countries, produce startups that truly scale, we built a bubble chart using color where darker blue signals higher average Year 3 revenue, and lighter blue signals lower average

Year 3 revenue. We specifically chose Year 3 revenue because it's the critical inflection point which tells us whether a startup successfully scaled past early growth, or simply plateaued. The three darkest bubbles: UK, South Korea, and France lead on average Year 3 revenue which means that startups in these countries didn't just survive, they scaled. The reason comes down to three structural advantages: strong government-backed funding that keeps startups alive through the difficult early years, deep research and development investment that builds defensible, hard-to-copy products, and mature corporate ecosystems that co-invest and provide clear exit pathways. For a venture capitalist, this is directly actionable because startups operating in or targeting UK, South Korea, and France carry a structurally higher probability of scaling past Year 3, which is where returns are generated.

## Visualization 4: Line Chart

**Focus:** Median Percentage Growth Rate by Launch Year and Current Status.

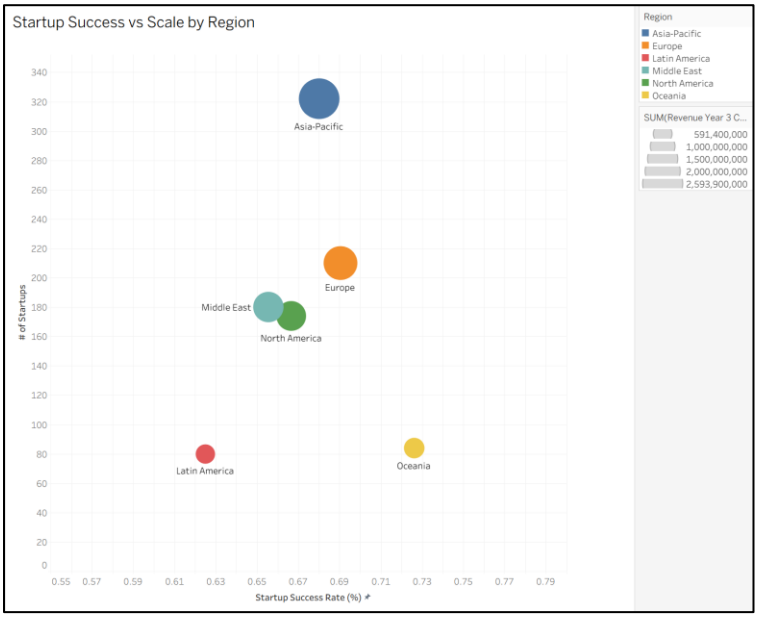



This Tableau visualization presents a cohort-based view of short-term startup revenue growth by plotting the median Year 1 to Year 2 growth percentage (Growth_pct) across launch years, segmented by each firm's Current Status (Successful, Acquired, and Failed). The horizontal axis represents the startup's Launch Date (year), while the vertical axis reports the median Growth_pct for all startups launched in that year within each status category, allowing the analysis to compare "typical" early scaling behavior while reducing distortion from extreme outliers that are common in startup financial data. By layering the three status-based trend lines on the same time scale, the visualization highlights meaningful differences in both growth potential and volatility across outcome groups: successful startups generally exhibit higher and more consistent median short-term growth over time, acquired startups display comparatively stable performance with greater year-to-year variation, and failed startups show the least stable growth pattern with pronounced fluctuations across cohorts. The year-by-year structure provides additional context for interpreting how market timing and broader business conditions may coincide with changes in early revenue scaling within this dataset, while the status segmentation

offers a practical lens for venture capital decision-making by connecting observed short-term growth trajectories to investment upside (higher growth) and risk (greater volatility).

## Visualization 5: Scatter Plot

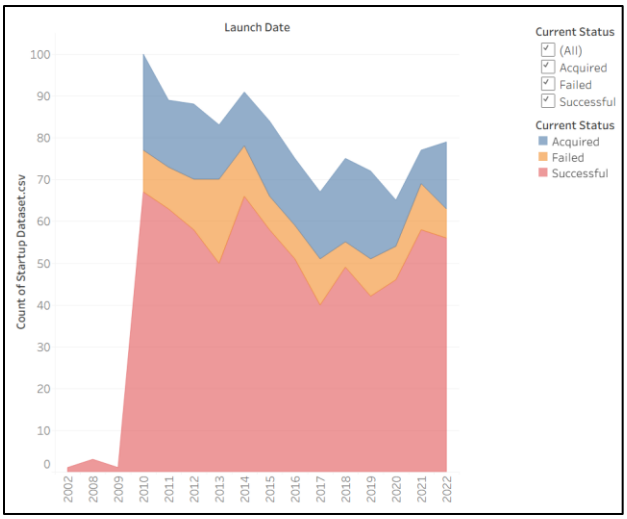
**Focus:** Startup Success vs Scale by Region.




To capture regional startup performance, a bubble chart showing startup success rate relative to the number of startups by region was created. Regional performance is important to evaluate as it helps clients understand not only where startup activity is concentrated, but where ecosystems may produce better results. This visualization compares the scale of each region's startup landscape through startup count while also measuring quality through success rate, with the bubble size representing the Year 3 Revenue variable. This allows users to identify regions that combine high startup volume, strong success rates, and revenue generation. Asia Pacific emerged as the strongest market having both the most startups and the largest revenue sum. Latin America appears to lag in both scale and performance, with this potentially showing an untapped market for a new startup.

## Visualization 6: Area Chart

**Focus:** Startup Launch Dates by Current Status.




To visualize startup launch date by current status, we created a stacked area chart that shows the count of startups from 2002-2022 by Acquired (blue), Failed (orange), and Successful (red). It is interesting to note launch date status from 2013. Successful startups, while still contributing the most, had its first steeper drop. Failed startups remained steady and acquired only fell slightly. This shows that, given a down year in successful startups, failed startups may run a bit steadier. During down-periods such as 2016-2020, it can also be noted that acquired startups had a much higher share of total startups than previous ranges. This is due to a steady decline in prior years where the total count of startups were measured. Along with that, more acquisitions were being completed due to a steady decline in successful startups.

# Conclusion

To conclude, the key startup success indicators are: High Average Year 3 Revenue, High Median Growth Rates, Low Volatility, High Founder Credibility, Business Confidence and Deep Talent Pipelines, Stable to High Economic Activity in a Country, Strong Government-Backed Funding, High Research and Development Investment and Strong Corporate Ecosystems.

Some of the other indicators that we could explore in the future are industry/sector, proximity to investor hubs, and exit strategy clarity. This is because deep tech and business-to-business software as a service (B2B Saas) consistently outperforms consumer-facing models in year 3 revenue. Then, startups with major metro ecosystems such as London, Seoul, and Paris can access capital faster. And lastly, startups with defined mergers and acquisitions (M&A) or initial public offerings (IPO) attract higher quality late-stage funding.

Together, these indicators provide a data-driven framework for Venture Capitalists to identify high-ROI startup investments.